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Q.5 a) In EM, we maximize a lower bound on the log-likelihood

In GEM, we take a step that increases this lower bound

As the ~~updt~~ update is

$$\theta^{(t+1)} = \theta^{(t)} + \alpha \nabla_\theta Q(\theta)$$

and α is small enough, the objective dont decrease. So lower bound increases every step.

As $l(\theta) > Q(\theta)$,

& increasing $Q(\theta)$ means

$$l(\theta^{(t+1)}) \geq l(\theta^{(t)})$$

Hence, the likelihood is monotonically increasing, so GEM converges.

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b) Log-likelihood

$$l(\theta) = \sum_i \log \sum_{z^{(i)}} p(x^{(i)}, z^{(i)}; \theta)$$

Taking gradient,

$$\nabla_\theta l(\theta) = \sum_i \frac{1}{\sum p(x^{(i)}, z^{(i)}; \theta)} \sum_{z^{(i)}} \nabla_\theta p(x^{(i)}, z^{(i)}; \theta)$$

This can be written as,

$$\sum_i \sum_{z^{(i)}} p(z^{(i)} | x^{(i)}; \theta) \nabla_\theta \log p(x^{(i)}, z^{(i)}; \theta)$$

Which is the same form as GEM update.